Remote Attestation of Devices with Redfish and SPDM

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Redfish

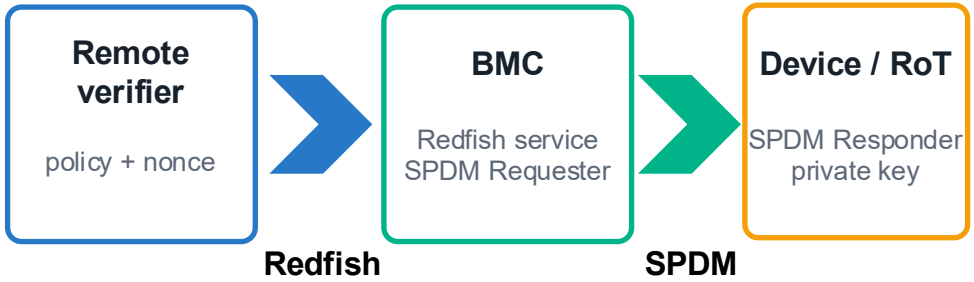
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Agenda

- Background
- SPDM Measurements Signing
- Redfish ComponentIntegrity Model
- SPDMGetSignedMeasurements Action
- Example End-to-End Attestation

Why Remote Attestation Matters

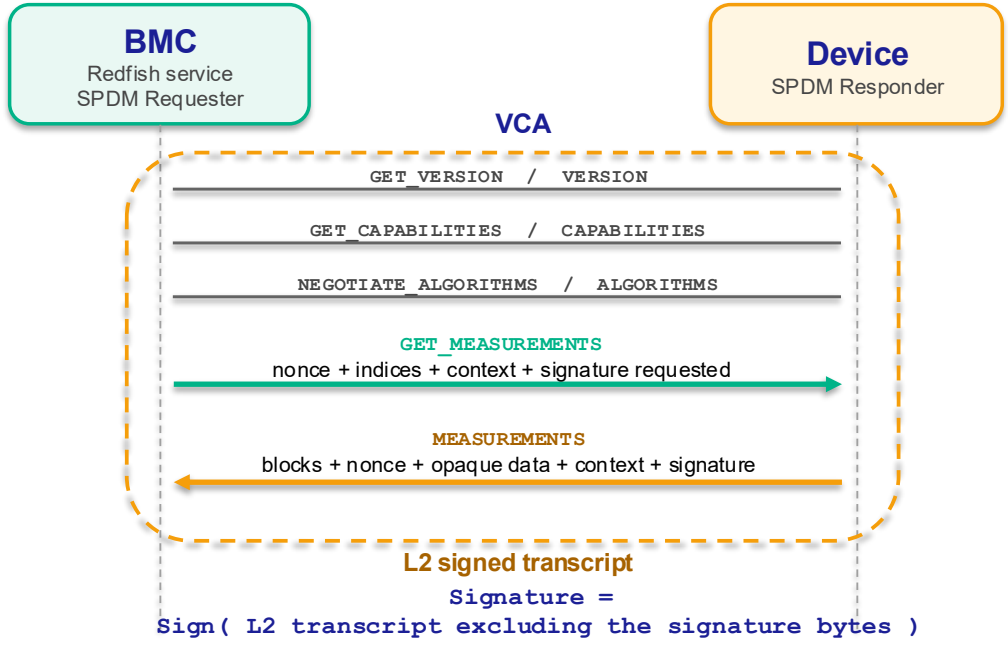
- BMC trust varies
  - Some BMCs have a hardware Root of Trust, others rely on software or a different trust anchor
- TPM placement varies
  - A TPM can attest the BMC but may sit outside the BMC's trust boundary or may not cover sufficiently
- Platform trust is not universal
  - A remote tenant or cloud operator may deliberately avoid trusting BIOS, host software, or BMC FW



nonce → fresh measurements → device signature

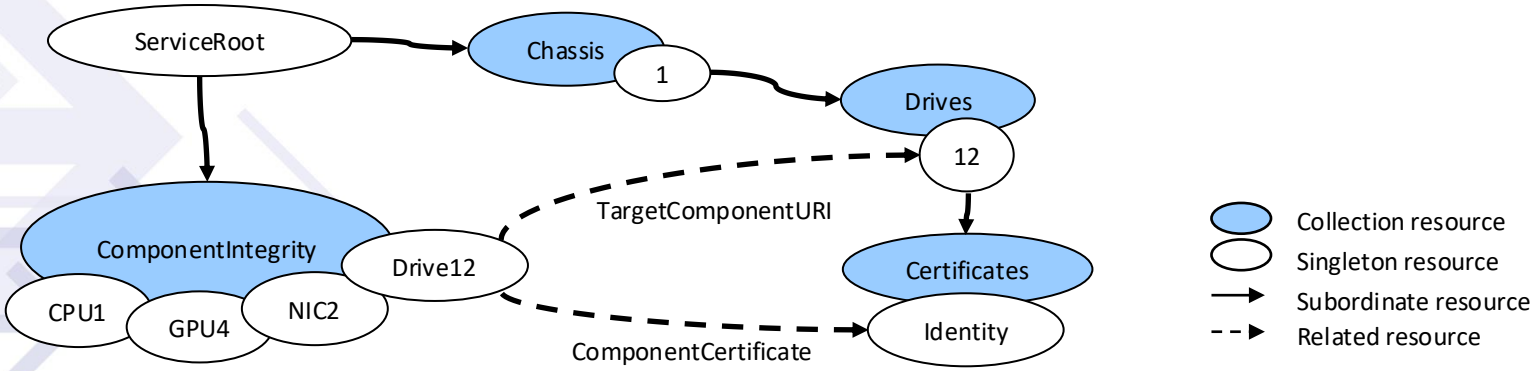
SPDM Measurements Signing

- Need to compute VCA
  - GET_VER/VERSION, GET_CAPS/CAPS, NEG_ALG/ALGORITHMS
- GET_MEASUREMENTS
  - Needs a nonce, measurement indices, and request a signature
- MEASUREMENTS
  - Includes the blocks, nonce, opaque data, and signature
  - Signature is computed over VCA, GET_MEASUREMENTS, and the MEASUREMENTS response excluding signature bytes using the signing algorithm negotiated in NEG_ALG/ALG



Redfish ComponentIntegrity Model

- The ComponentIntegrity resource contains security-related information between two devices
  - Typically, a BMC and a managed device
  - In SPDM, this maps to the SPDM Requester and SPDM Responder



ComponentIntegrity Mockup


```
{
  "@odata.type": "#ComponentIntegrity.v1_5_0.ComponentIntegrity",
  "Id": "Drive12",
  "Name": "Drive 12 Component Integrity",
  "Status": { "Health": "OK", "State": "Enabled" },
  "ComponentIntegrityType": "SPDM",
  "ComponentIntegrityTypeVersion": "1.3.0",
  "ComponentIntegrityEnabled": true,
  "TargetComponentURI": "/redfish/v1/Chassis/1/Drives/12",
  "LastUpdated": "2026-08-15T14:05:39-04:00",
  "SPDM": {
    "Requester": { "@odata.id": "/redfish/v1/Managers/BMC" },
    "MeasurementSet": {
      "MeasurementSpecification": "DMTF",
      "MeasurementSummary": "oylHo8JAT1RqtB0Vlx9dCzogavikhSvIZpVEDhI1X1+TKeyZZvwuUb5qF9uWqPVa",
      "MeasurementSummaryHashAlgorithm": "TPM_ALG_SHA_384",
      ...
    },
    ...
```


Basic Info and Status

SPDM-specific Info

Measurement Data

ComponentIntegrity Mockup

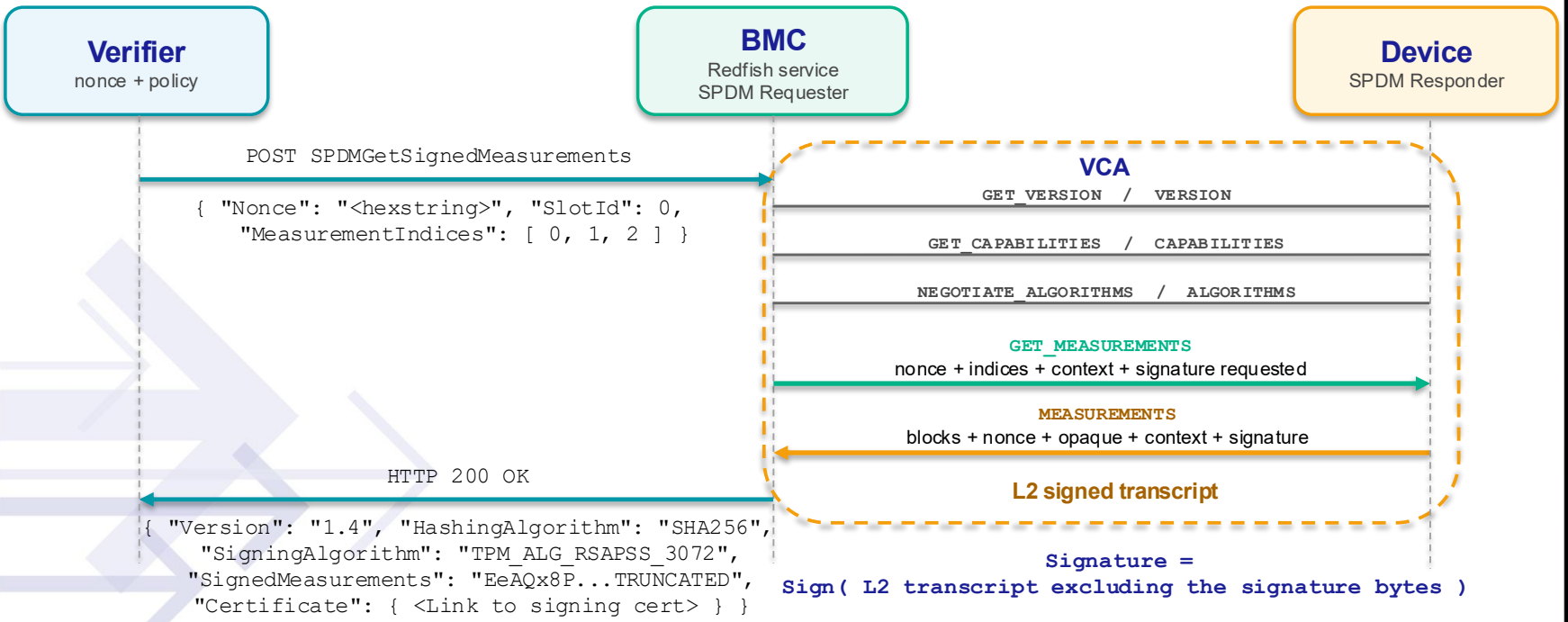
```
    ...
    "IdentityAuthentication": {
      "ResponderAuthentication": {
        "ComponentCertificate": { "@odata.id": "/redfish/v1/Chassis/1/Drives/12/Identity" },
        "VerificationStatus": "Success"
      }
    },
    "ComponentCommunication": {
      "Sessions": [
        { "SessionId": 15999129, "SessionType": "EncryptedAuthenticated",
          "SessionAlgorithm": "TPM_ALG_ECDSA_ECC_NIST_P521" }
      ]
    }
  },
  "Actions": {
    "#ComponentIntegrity.SPDMGetSignedMeasurements": {
      "target": "/redfish/v1/.../Actions/ComponentIntegrity.SPDMGetSignedMeasurements"
    }
  }
}
```

Authentication Info

Session Info

Actions

SPDMGetSignedMeasurements Action



Example End-to-End Attestation

- DMTF's Redfish Tacklebox project contains a script to perform attestation with the SPDMGetSignedMeasurements action
  - rf_component_integrity.py
  - Subcommand "getsignedmeasurements"
- The script…
  - Finds the requested ComponentIntegrity resource
  - Invokes SPDMGetSignedMeasurements with a random nonce
  - Parses the L2 log from the action response
  - Verifies the nonce in the L2 log matches the provided random nonce
  - Finds and verifies the certificate chain used to sign the measurements
  - Verifies the signature in the L2 log the public key in the certificate chain

rf_component_integrity.py Example (List)

```
$ rf_component_integrity.py -u root -p root -r https://10.2.3.50
  Id                | Details
  CPU1              | CPU 1 Component Integrity
                    | Component: /redfish/v1/System/1/Processors/1
                    | Type: SPDM; State: Enabled/OK
  GPU4              | GPU 4 Component Integrity
                    | Component: /redfish/v1/System/1/Processors/GPU4
                    | Type: SPDM; State: Enabled/OK
  NIC2              | Network Adapter 2 Component Integrity
                    | Component: /redfish/v1/Chassis/1/NetworkAdapters/2
                    | Type: SPDM; State: Enabled/OK
  Drive12           | Drive 12 Component Integrity
                    | Component: /redfish/v1/Chassis/1/Drives/12
                    | Type: SPDM; State: Enabled/OK
```

rf_component_integrity.py Example (Measurements)

```
$ rf_component_integrity.py -u root -p root -r https://10.2.3.50 getsignedmeasurements -ci Drive12 -v
Signed Measurements:
  Type: SPDM
  SPDM Version: 1.3.0
  Signing Algorithm: TPM_ALG_ECDSA_ECC_NIST_P384
  Hashing Algorithm: TPM_ALG_SHA_384

  Measurements:
    Index: 1, Specification: 01
    Data: 83...01
    Index: 2, Specification: 01
    Data: 82...ff
    ...
    Index: 51, Specification: 01
    Data: 82...397d84e706cd
    Index: 52, Specification: 01
    Data: 81...02010200f121
```

Call to Action

- Try the Redfish Use Case Checkers!
- Bugs, questions, feature requests?
  - File an issue on GitHub, or make a pull request
  - The Redfish Forum reviews issues and pull requests during meetings
- GitHub links
  - https://github.com/DMTF/Redfish-Tacklebox

Thank you!

For more information,
visit us online at
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Redfish Developers Hub at
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